

Zachary Zeppieri

Senior Continuous Improvement Engineer · Manufacturing Systems & Operations Excellence

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SUMMARY

- Senior continuous improvement leader and sole CI practitioner at a Tier-1 aerospace MRO through workforce reductions; promoted twice in three years and trusted with a multi-million-dollar Manufacturing Execution System (MES) project deployment.
- Delivered 30,000+ hours of verified productivity savings, 29 Kaizen events at 5 sites across the US, 77+ employees trained, and deployed self designed 3D printed solutions across the plant.
- Ship the production tooling the floor needs — Python desktop apps, a custom Tulip MES framework, and a patent-pending engineered fixture. Six Sigma Green Belt; B.S. Mechanical Engineering.

EXPERIENCE

Chromalloy Gas Turbine, LLC | *Senior CI Analyst* (2025–Present) · *CI Analyst* (2023–2025) · *Associate, CI Analyst* (2022)

- Reduced Blade Cell Gate 1 turnaround time (TAT) from 14 days to 4.75 days against a 4-day target — designed the Line of Balance board, standardized Op20 induction, and led the gate 1 improvement Kaizen activities.
- Drove a 5S Go-To-Gemba program from a 15-point baseline to 85 points by end of 2024 — 16 Bronze, 17 Silver, and 3 Gold-rated areas — maintained with three-shift coaching and a leadership Kamishibai accountability board.
- Delivered ~30,000 hours of verified productivity savings across all product lines (2024–2025) — exceeded the 2025 18.1K-hour target by 30% across 80 candidate projects.
- Designed and ran a Continuous Improvement Lunch-and-Learn program — 14 sessions of 2-Second Lean and Lean Lego Game training reached 77+ recorded employees across all three shifts, building lean fluency from the operator layer up.
- Facilitated 29 Kaizen events across five Chromalloy sites (2024–2025), including ten external multi-day cross-functional engagements — a CFM56-7B 3P design event, Plasma / Laser Drilling / Heat Treat TPMs, and the CSD Lean Leadership Week engine bay overhaul.
- Lead the Chromalloy customer side of a multi-million-dollar Tulip MES deployment at the **Windsor (CT) site** with PwC as implementation partner — driving Project Plan, Validation Approach, Base Layout, and core architecture decisions toward a May 2026 pilot and September 2026 go-live.
- Built a Microsoft Forms + Power Automate system that captures Kaizen and 2-Second Lean improvements network-wide — 211 corporate entries with zero ongoing maintenance; in use across Chromalloy.
- Mentored a continuous-improvement intern who was hired full-time as an engineer; Expecting to manage 1-2 full time CI Practitioners Q4 26
- Sole CI practitioner at the New York site through a multi-year workforce reduction — sustained and grew the lean program's measurable output during a contracted-headcount period.

PROJECTS

Blade Cell Capital Project Management | *Chromalloy New York* — \$2.4M, team of 20 Nov 2022 – Jun 2023

- Led a \$2.4M capital project from CapEx approval (Nov 2022) through deployment — directed a team of 20, applied 3P / cellular-flow design, coordinated purchasing, finance, engineering, and operations, and returned \$350K to budget for critical spares.

Tulip MES Site Deployment | *Windsor, CT* — *AI-augmented program management* Feb 2026 – Present

- Authored the program governance backbone — the RACI / escalation path, project management framework, and the risk register driving weekly RAID reviews; contributed to the multi-million-dollar business case (multi-hundred-percent 8-year ROI, ~1-year payback) and the vendor commercial negotiations.
- Architected an AI-augmented PM toolchain that runs the in-flight program — a custom Claude-based project-management framework, an offline Obsidian-style D3 knowledge graph with prompt-engineered context retention across the full project corpus, and Microsoft Copilot automations that ingest project emails and Teams messages to keep the knowledge base current without manual entry.
- Built a Smartsheet writeback pipeline that pushes program status, RAID items, and milestone tracking from the project knowledge base into stakeholder viewed Smartsheet automatically — supporting bi-weekly executive updates without manual report assembly.

- Authored a supporting Python framework (**TulipAppBuilder**) that generates importable Tulip MES apps from build scripts, codifying 50+ reverse engineered platform rules to accelerate the rollout’s solution-specific deliverables; shipped six production apps including a 5-app MES Router suite (Master Router, work-instruction step player with signature capture, routing maintenance, master traveler, NCR reporting).

OEM Tracker / Master Planning Tool | *Python desktop + web, shipped* May 2026

- Replaced a 14 MB / 70-sheet VBA Excel macro tracker with a Python desktop app (PySide6 GUI, SQLite cache) distributed as a Windows executable — multi-user concurrent design, sub-second responsiveness across 25 OEM tabs and three plant views.
- Built a parallel Python web version using only the Python standard library — drag-and-drop CSV / XLSX import, scheduled auto-imports, KPI dashboard, 28-day shipping calendar — zero external dependencies.

Baffle Extraction Tool (Patent Pending) | *Chromalloy New York* Jul 2024 – Sep 2024

- Designed and developed a patent-pending fixture enabling in-house baffle refurbishment instead of OEM purchase — estimated \$100K+ annual savings; currently in patent application.

Reporting Automation & 3D-Printed Shop-Floor Engineering | *Chromalloy New York* 2024 – Present

- Automated 90% of monthly productivity reporting (4+ hours → under 30 minutes) and standardized AI-assisted Value Stream Map generation (>40 hours → near zero) for ~100 operations.
- Designed dozens of 3D-printed fixtures deployed on the production floor — including CBN boot molds delivered in under 24 hours that saved 2–3 days of vendor TAT for 59 repair blades, a polishing-disc contamination stand, and an LM9000 plasma-tape escape alignment tool.

SKILLS

Lean & Continuous Improvement (multi-site): Six Sigma (Green Belt), Kaizen / K3 facilitation (29 events, 5 sites), Value Stream Mapping, Total Productive Maintenance, 3P, 5S+1, Toyota Production System, 2-Second Lean, Standard Work, Line of Balance, Kamishibai, T3/T5 tier boards

Manufacturing Execution Systems: Tulip MES (PM + custom app development), MES Router Framework design, work-instruction digitization, SyteLine ERP integration, traveler design

Program & Project Management: Multi-million-dollar capital and software programs, RACI design, risk register, weekly RAID reviews, vendor / partner commercials (PwC, Tulip), cross-functional steering committees

Coaching, Training & Change Management: Lunch-and-Learn program design (100+ trained), front-line operator coaching, leadership accountability, employee-ideas program (422 ideas implemented in 2025), people leadership (intern → full-time hire)

Software & Automation Tooling (AI-augmented): Python (PySide6, SQLite), JavaScript (vanilla, D3.js), HTML / CSS, Microsoft Power Automate + Forms, AI co-pilot integration (Claude, Copilot)

Domain: Aerospace MRO, Gas Turbine Engine Repair, Regulated Industry

CAD & Engineering: Fusion, Siemens NX, Visio, SketchUp, 3D printing for rapid shop-floor tooling, GD&T

EDUCATION

State University of New York at New Paltz

New Paltz, NY

Bachelor of Science in Mechanical Engineering

2022

CERTIFICATIONS & RECOGNITION

- Six Sigma Green Belt (Chromalloy, 2023)
- Patent Pending — Baffle Extraction Tool (filed 2024)
- 4 Strong / High Achiever performance rating, 4 consecutive cycles; two promotions in 3 years
- Cross-site lean leadership at Chromalloy sister facilities — Arizona, Holland, San Diego, Nevada, Florida, Connecticut